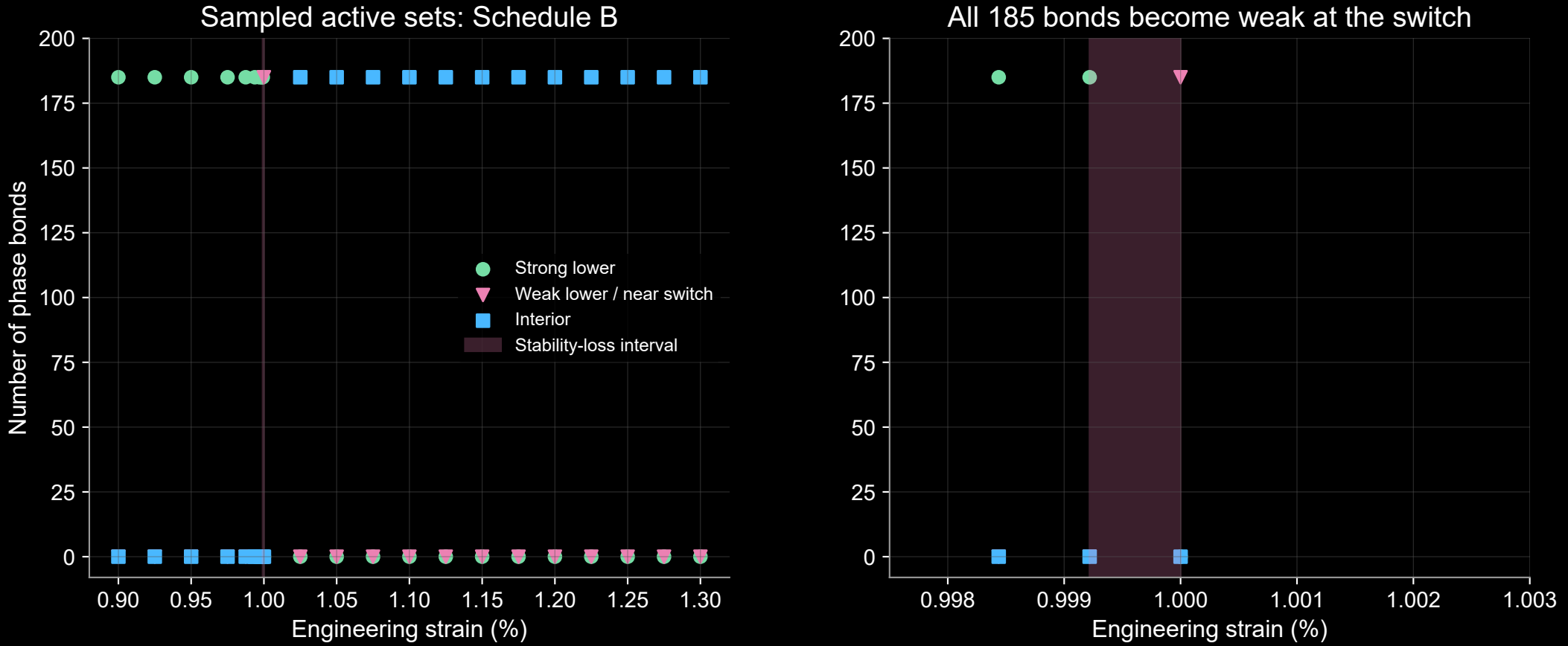

Strong lower bounds prevent phase opening in the critical cone
Weak lower bounds permit one-sided phase opening; sampled interior states occur above 1.00%